Trinity Engine Canon Review Report

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1 Executive Assessment

Question	Assessment
Is the canon complete enough to support implementation without interpretation?	Yes, A1..A
Does ambiguity, conflict, or missing implementation-critical detail remain?	No (canon-
Is the package sufficient to begin deterministic implementation under the agreed scope?	Yes — and

- High-level judgment (2026-04-28 revision):
  - All seven critical/significant ambiguities raised against the 2026-03-24 review have been ruled on by the canon owner and folded back into the governing documents.
  - One ruling (A3) corrected the implementation's working assumption (midpoint selection) to the canonical rule (lower-bound selection); the solver and the $\text{design}_{\text{timeutc}}$ oracle have been updated accordingly.
  - The remaining open work is operational, not canonical: golden fixture population, baseline approval, and acceptance-review execution.

2 Strengths Of The Package

- Scope is explicit and narrow.
  - The package is disciplined about what is in scope and what is excluded.
  - The engine boundary is clear: normalized input only.
- Core domain mechanics are substantially pinned.
  - Human Design gate order, anchor, line width, center list, channel table, Type rules, and Authority priority are all explicitly formalized.
  - Astrology source stack, supported objects, sign mapping, and output structure are explicit.
- Determinism governance is strong.

  - Canon precedence, versioning, regression obligations, and no-hidden-state rules are unusually well documented.

- Transparency is good.
  - Document 09 and Document 12 make the decision history and prior rejected assumptions visible, which materially reduces hidden-context risk.

3 Critical Ambiguities And Blockers (RESOLVED 2026-04-28)

3.1 A1. Timezone Source Version — RESOLVED (D20)

- Documents (folded back):
  - Document 03 §3.2 — pinned IANA tzdb release
  - Document 08 §4.3 and §6.4 — source-stack pin requirement
  - Document 12 D20
- Ruling:
  - The authoritative IANA Time Zone Database release for Trinity Engine v1 is **2026a**.
  - The implementation MUST NOT rely on the tzdata embedded in the Go runtime unless that embedded release is confirmed to match 2026a.
  - Runtime or operating-system default timezone data is non-authoritative wherever it can affect canonical UTC conversion.
- Implementation status:
  - `pkg/canon/sources.go: TZDBVersion` updated from working-assumption `2023c` to canon `2026a`.
  - The build still imports `time/tzdata` for portability; an infrastructure follow-up will vendor IANA 2026a explicitly (e.g. a generated tzdata package or sidecar $\mathrm{TZ}_{\mathrm{DIR}}$ mount) and add a startup assertion verifying a 2026a-sentinel DST rule.

3.2 A2. House Interval Ownership — RESOLVED (D21)

- Documents (folded back):
  - Document 05 §5.6 — house placement rule
  - Document 06 §3.4 — boundary semantics
  - Document 12 D21
- Ruling:
  - House intervals are **cusp-inclusive** and **next-cusp-exclusive**.
  - A longitude exactly equal to cusp N belongs to House N; a longitude exactly equal to the next cusp belongs to the next house.
  - The House 12 → House 1 wrap follows the same rule.
  - No epsilon, midpoint, dual-inclusive, or library-default ownership rule is allowed.
- Implementation status:
  - `pkg/trinity/astro/house.go::HouseFor` already implements this exactly. Comments updated from "working assumption" to RESOLVED.

3.3 A3. Design-Time Final Timestamp Selection — RESOLVED (D22, implementation corrected)

- Documents (folded back):
  - Document 05 §6.4.5–6.4.7 — bisection stop conditions
  - Document 05 §11.2 — emission rule
  - Document 07 §10.3–10.4 — output formatting
  - Document 10 — fixture exactness
  - Document 12 D22
- Ruling:
  - When the design-time bisection stops because the remaining interval width is < 1 second, the canonical Design time is the **lower bound** of the final search interval.
  - The emitted `design_time_utc` uses whole-second precision by **truncation** of that lower bound.

  - No midpoint, nearest-second, ceiling, or alternate adjustment is allowed.

- Implementation status:
  - The previous implementation returned the bracket midpoint at the width-stop. `pkg/hd/calc/designtime.go` has been corrected: the width-stop and the (pathological) max-iterations exit now both return `lower`.
  - `output.DesignTime.MarshalJSON` already truncates to whole seconds and is unchanged.
  - The Schiedam `design_time` oracle may shift by up to one whole second under the corrected solver; the regression sentinel will need re-baseline if the truncated value crosses a second boundary.

3.4 A4. Error Message Comparison — RESOLVED (D23)

- Documents (folded back):
  - Document 07 §9.5
  - Document 10 §6.3
  - Document 12 D23
- Ruling:
  - Canonical error fixture comparison is based on `error_type` and `error_code` (plus required envelope shape and metadata).
  - The `message` field must be present where required, but its exact wording is not authoritative for fixture exactness unless a later canon revision introduces a formal message catalogue.
- Implementation status:
  - The validator, error envelope, and golden runner already implement type-only comparison with non-empty message. Comments updated to RESOLVED.

4 Significant Ambiguities (RESOLVED 2026-04-28)

4.1 A5. `invalid_input` vs `unsupported_input` Boundary — RESOLVED (D24)

- Documents (folded back):

  - Document 04 §6.2 and §11
  - Document 07 §9.4
  - Document 10 §9.3
  - Document 12 D24

- Ruling:
  - Malformed payloads or values that violate the canonical input contract are `invalid_input`.
  - Structurally valid payloads that fall outside v1 supported scope are `unsupported_input`.
  - The v1 input boundary is strict: minute-precision birth time, one person per payload, no optional canonical input fields, no silent coercion / repair / normalisation inside the engine.
  - Multi-person payloads are `unsupported_input`.

- Implementation status:
  - `pkg/trinity/input/validator.go` already classifies on this boundary. Comments updated to RESOLVED.

4.2 A6. IANA Link / Alias Names — RESOLVED (D24)

- Documents (folded back):
  - Document 04 §6.3
  - Document 03 §3.2
  - Document 12 D24

- Ruling:
  - Accepted timezone identifiers are restricted to canonical IANA Area/Location identifiers from the pinned tzdb release.
  - Aliases and link names (e.g. `US/Eastern`) are not accepted unless introduced by a later canon revision.
  - No silent canonicalisation occurs inside the engine. Upstream systems may canonicalise inputs before calling, but that is outside the engine boundary.

- Implementation status:

  - The validator already rejects abbreviations, the documented link prefixes, and any identifier `time.LoadLocation` cannot resolve. Comments updated to RESOLVED. A follow-up will replace the prefix list with a curated zone.tab when the vendored tzdata 2026a is available.

4.3 A7. Golden Fixture Bootstrap — RESOLVED (D26)

- Documents (folded back):
  - Document 10 Appendix A
  - Document 12 D26
- Ruling:
  - The first golden fixture set must not become self-validating.
  - Candidate expected results may be generated from a canon-compliant implementation, but generation by the implementation is necessary and **not sufficient**.
  - Before approval, fixtures must be reviewed against the governing canon — including independent review of ambiguity-prone areas: timezone-sensitive cases, sign / house / gate / line boundaries, design-time sub-second stop behaviour, error classification edge cases.
  - External tools may be used for sanity checking but are not the golden oracle.
- Implementation status:
  - All Phase-N regression sentinels (astrology, design-time, activations, structure, $\text{gene}_{\text{keys}}$) carry comments updated to reflect D26: cross-checks are sanity checks, not oracles; canon-owner approval is required before the freeze is authoritative.

4.4 A8. The Canon Itself Admits The Final Consistency Pass Is Not Fully Folded Back

- Documents:
  - Document 00
  - Document 09 item O02

- Problem:
  - The package still records a pending fold-back/completeness step for Documents 05-07.
  - That means identifier hygiene and cross-document wording drift remain known risks.
- Why this matters:
  - An engineer can proceed, but should expect some late canon edits.

5 Potential Conflict Or Tension Points

5.1 C1. "Byte For Byte" Exactness Versus Partially Unpinned Serialization

- Source context:
  - Earlier email language emphasizes exact expected output "byte for byte."
  - Output Specification focuses on canonical structure, field presence, ordered arrays, and pinned numeric/date formatting.
- Tension:
  - The package does not fully pin object member order or a canonical JSON serializer.
  - Document 07 explicitly allows some serialization variation such as trailing zeros depending on implementation serialization.
- Assessment:
  - If review is semantic JSON comparison, this is fine.
  - If review is literal byte-level comparison, this remains underspecified.

6 Readiness By Workstream (revised 2026-04-28)

Workstream	Readiness	Notes
Input validation	Ready	Strongest part of the can
Source stack setup	Ready (with infra follow-up)	Swiss Ephemeris 2.10.03
Astrology signs and object ordering	Ready	Sign mapping and object
Astrology house placement	Ready	A2/D21 pins cusp-inclusi
Human Design gate/line mapping	Ready	Gate order, widths, ancho
Human Design structural derivation	Ready	Channels, centers, definit
Human Design design-time emission	Ready (oracle re-baseline pending)	A3/D22 lower-bound rule
Gene Keys derivation	Ready	Scope is narrow and map
Error handling	Ready	A4/D23: $\text{error}_{\text{type}}$-only c
Golden test scaffolding	Ready	Structure and coverage a
Golden fixture truth establishment	Pending approval	A7/D26: engine generati

7 Overall Conclusion (2026-04-28)

- All seven canon-level ambiguities raised in the 2026-03-24 review (A1..A7) are now RESOLVED and folded back into the governing canon.
- The implementation has been updated to match every ruling. The only behavioural correction was A3 (midpoint → lower-bound at the design-time width-stop); all other rulings ratified the working assumptions already implemented.
- C1 ("byte-for-byte" vs serialisation) is resolved by D25: comparison is **semantic JSON equality** under canon-pinned formatting and ordering rules, not literal byte equality.
- A8 (final consistency pass across Documents 05/06/07) is closed by Document 00's "recently folded back closure decisions" list (2026-03-31 update).
- Remaining work is operational, not canonical:
  - re-baseline the Schiedam `design_time` oracle if the A3 fix shifts the truncated whole-second value
  - populate the golden fixture set per Document 10 minimums
  - obtain canon-owner approval for the frozen baseline (D26)
  - execute the acceptance review per Document 11 Appendix A

8 Resolved Ambiguities Tracker

Tag	Status	Canon Decision	In
A1	RESOLVED 2026-04-28	D20 — IANA tzdb 2026a pinned	`ca`
A2	RESOLVED 2026-04-28	D21 — cusp-inclusive / next-cusp-exclusive	`as`
A3	RESOLVED 2026-04-28	D22 — lower-bound + truncation	`ho`
A4	RESOLVED 2026-04-28	D23 — $\text{error}_{\text{type}}$ / $\text{error}_{\text{code}}$ authoritative; message informational	go
A5	RESOLVED 2026-04-28	D24 — strict invalid vs unsupported boundary	`in`
A6	RESOLVED 2026-04-28	D24 — canonical IANA Area/Location only	`in`
A7	RESOLVED 2026-04-28	D26 — generation necessary, approval required	se
A8	RESOLVED 2026-03-31	Document 00 closure list	id
C1	RESOLVED 2026-04-28	D25 — semantic JSON equality	`g`

9 Recommended Next Step (2026-04-28)

- Re-run the engine integration suite under the corrected A3 rule and re-baseline `baseline/schiedam_1990_04_09_design_time.json` if its whole-second value moved.
- Vendor IANA tzdb 2026a explicitly into the build (separate infra task).
- Populate remaining golden fixture categories (Document 10 minimums) and submit the baseline for canon-owner approval per D26.
- Execute the Document 11 Appendix A acceptance-review checklist.